

Taming the MLOps Zoo

Orchestrating & Monitoring Models with Airflow



Lindy Bustabad

Sr. Analytics Engineer, Pendo

Introducing Pendo

Your all-in-one Software Experience Management platform

GATHER

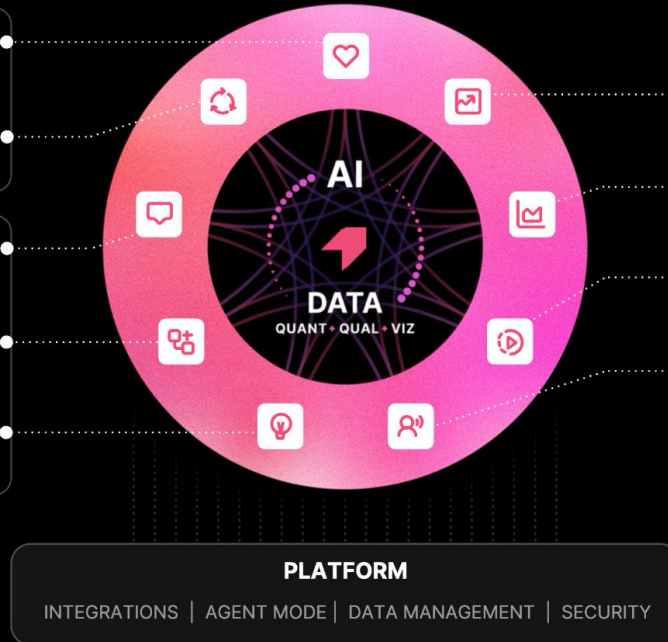
- Sentiment**
Understand how users feel
- Data Sync**
Export your Pendo data

- Guides**
Communicate with users
- Orchestrate**
Communicate across channels
- Predict**
Predict churn and upsell

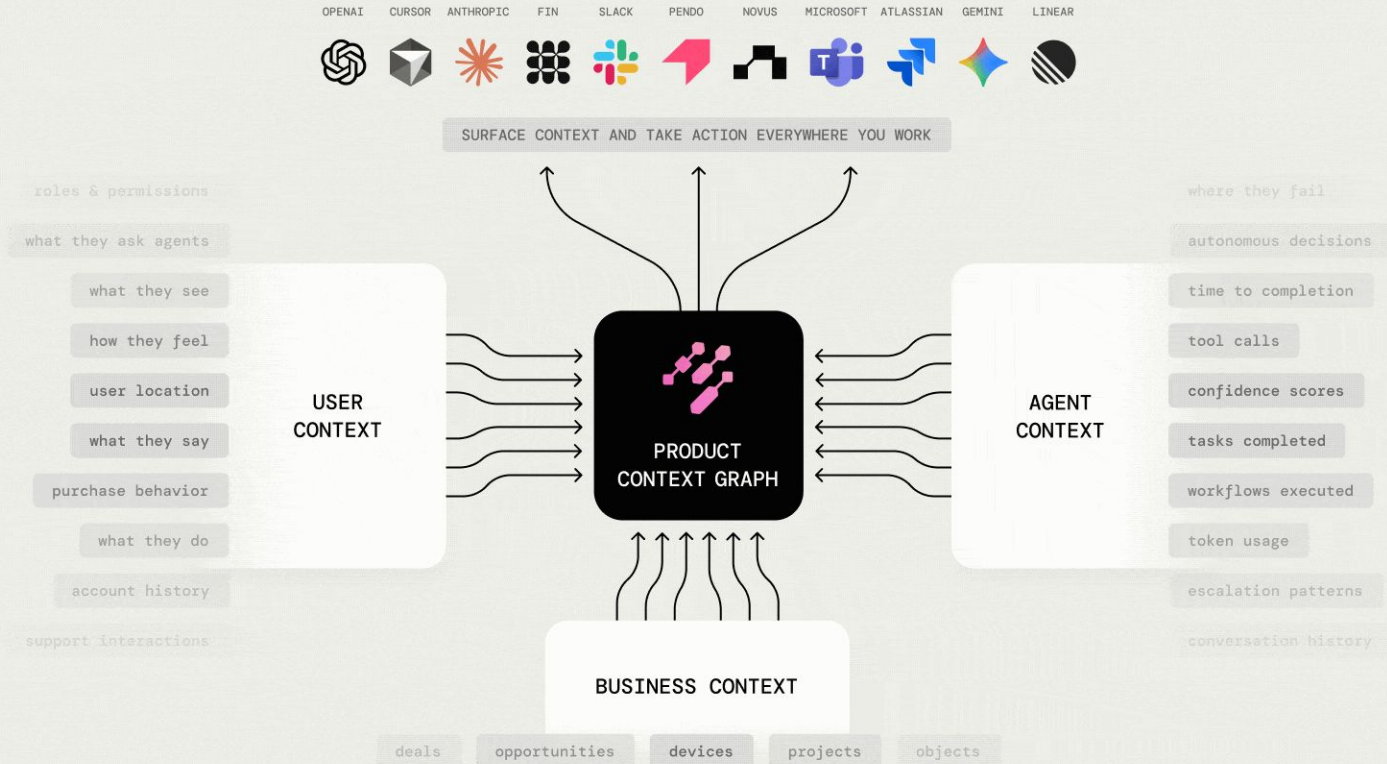
ACT

LEARN

- Product Analytics**
Understand what users do
- Agent Analytics**
Understand what agents do
- Session Replay**
Visualize user behavior
- Listen**
Build the right products



Pendo is the product context layer



**At Pendo, AI collapsed our
time-to-production from
3 months to 3 weeks.**



At Pendo, AI collapsed our
time-to-production from
3 months to 3 weeks.

The new bottleneck is **attention.**



 AI Ascent

Greg Brockman
/ OpenAI



0:00 / 28:26

Introduction >



OpenAI's Greg Brockman: Why Human Attention Is the New Bottleneck

ML ORCHESTRATION

The MLOps Zoo

Python ML on K8s



dbt / SQL Scorecards



BigQuery Native ML



Pendo Predict
(managed ML)



Hybrid



“Feeding Schedules”

Python ML on K8s



✓ 3x/day

dbt / SQL Scorecards



✓ Weekdays

BigQuery Native ML



✓ Monthly

Pendo Predict
(managed ML)



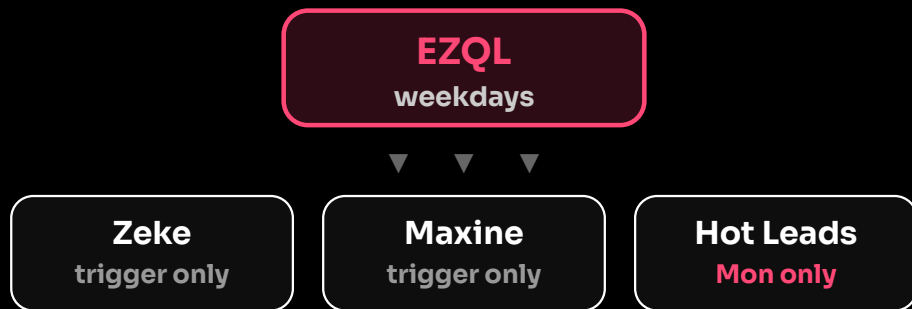
✓ Daily

Hybrid



✓ Daily

DAG Ecosystem



Independent DAGs



How We Feed the Animals

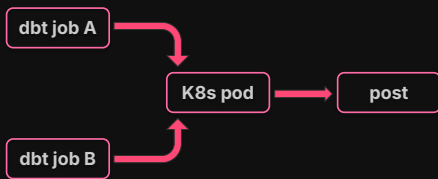


GRAPH VARIATIONS

How We Feed the Animals

Parallel Prep, Single Scorer

A/B model versions share one scoring container



fan-in before scoring · fan-out after



How We Feed the Animals

Parallel Prep, Single Scorer

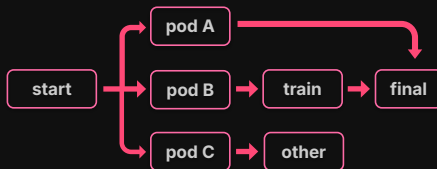
A/B model versions share one scoring container



fan-in before scoring · fan-out after

Partial Fan-In

Not all downstream tasks wait for all upstream



trigger_rule='none_failed'

How We Feed the Animals

Parallel Prep, Single Scorer

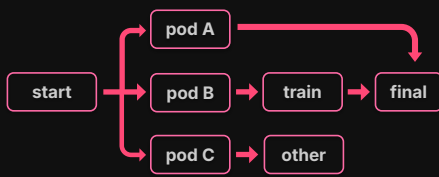
A/B model versions share one scoring container



fan-in before scoring · fan-out after

Partial Fan-In

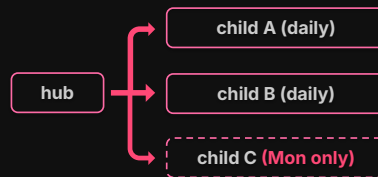
Not all downstream tasks wait for all upstream



trigger_rule='none_failed'

Conditional Child DAGs

Skip expensive branches by schedule



BranchPythonOperator +
TriggerDagRunOperator



>30

Daily dbt jobs

~10

K8s pods spun up

1-3

Scoring rounds
per model /day

... but which enclosures need a **zookeeper**?



THE ZOOKEEPER

DS ModelGuard

Separate Airflow DAG, runs daily

Checks: Precision/recall vs. thresholds,
volume anomalies, feature drift (PSI, z-scores),
dbt upstream health

Output: Slack health report, per-model status



DS ModelGuard APP 9:29 PM

Pendo ML Daily Health Report

Monday, August 31, 2026

Charlie Churn Prediction

- Upstream dbt Health
- Charlie Churn Prediction: Charlie - Scoring V1: succeeded Aug 31, 6:05 AM ET
- 1 - Daily: succeeded Aug 31, 6:31 AM ET
- 1 - Daily - Pendo Usage: succeeded Aug 31, 4:29 AM ET

Volume: 469 | Score: 0.4085

Ward Segment B

- Upstream dbt Health
- Ward Segment B: Predict Base: succeeded Aug 31, 9:00 AM ET
- 1 - Daily: succeeded Aug 31, 6:27 AM ET
- 1 - Daily - Pendo Usage: succeeded Aug 31, 8:34 AM ET

Accuracy

- Precision: 0.5595 (threshold: 0.33)
- Recall: 0.5425 (threshold: 0.5)
- Volume: 170 accounts scored

Ward Segment A

- Upstream dbt Health
- Ward Segment A: Predict Base: succeeded Aug 31, 9:00 AM ET
- 1 - Daily: succeeded Aug 31, 6:48 AM ET
- 1 - Daily - Pendo Usage: succeeded Aug 31, 9:31 AM ET

Accuracy

- Precision: 0.4567 (threshold: 0.51)
- Recall: 0.4863 (threshold: 0.6)
- Volume: 612 accounts scored

Detection is solved.
Prioritization is not.



Detection is solved. Prioritization is not.

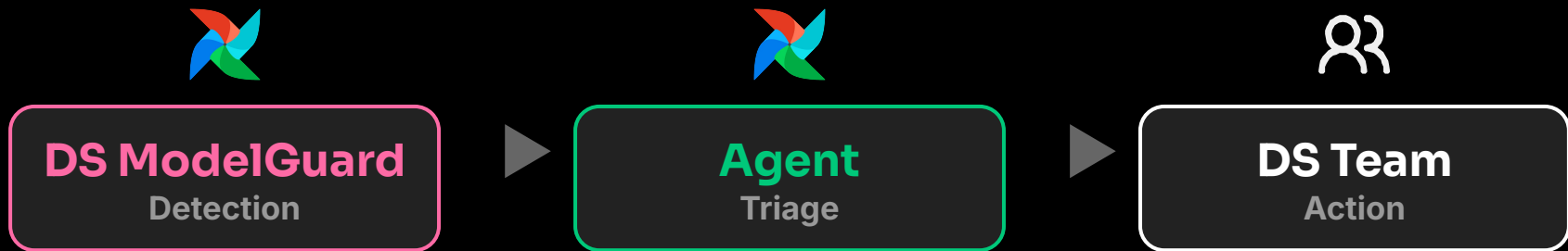
Which yellow status matters more?

Is this the same pattern we saw last month?

Where should I start?



What If an Agent Did the Rounds?



Choose which models to drill into

Compare today's issues against past incidents

Prioritize, "check Ward first, Charlie can wait"

pendolytics

All 11 Active 11 Paused 0

Running 1 Failed 0

x MLOps

Search DAGs

Auto-refresh

	DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks	Actions	Links
	charlie_dbtjobs MLOps	airflow	184	5 11,17,22 ***	2026-08-27, 17:05:00	2026-08-27, 22:05:00	4	...	
	eve_dbtjobs MLOps	airflow	1	0 12 1 * *	2026-07-01, 12:00:00	2026-08-01, 12:00:00	6	...	
	ezql_dbtjobs MLOps	airflow	43 2	55 11 * * 1-5	2026-08-26, 11:55:00	2026-08-27, 11:55:00	11 1	...	
	ezql_hot_leads_dbtjobs MLOps	airflow	11	None	2026-08-24, 11:56:21		3	...	
	ezql_maxine_dbtjobs MLOps	airflow	46 2	None	2026-08-27, 11:56:09		7	...	
	ezql_zeke_dbtjobs_ab MLOps	airflow	46	None	2026-08-27, 11:56:09		3	...	
	florence_dbtjobs MLOps	airflow	61	0 12 * * *	2026-08-26, 12:00:00	2026-08-27, 12:00:00	3	...	
	modelguard_daily_report MLOps	airflow	4	30 10 * * *	2026-08-26, 14:30:00	2026-08-27, 14:30:00	1	...	
	modelguard_triage MLOps	airflow	57	None	2026-08-27, 22:32:04		2	...	
	nightingale_dbtjobs MLOps	airflow	171 1 8	20 11,17,22 * * *	2026-08-27, 17:20:00	2026-08-27, 17:20:00	5 1 3	...	
	rodrigo_dbtjobs_ab MLOps	airflow	177 5	5 12,17,23 * * *	2026-08-27, 12:05:00	2026-08-27, 17:05:00	9	...	

pendolytics

All 11 Active 11 Paused 0

Running 1 Failed 0

× MLOps

Search DAGs

 Auto-refresh 

DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks	Actions	Links
 charlie_dbtjobs MLOps	airflow	 184 	5 11,17,22 *** 	2026-08-27, 17:05:00 	2026-08-27, 22:05:00 	            		



DS ModelGuard APP 10:30 AM

Today ▾



Pendo ML Daily Health Report



Thursday, August 27, 2026



Charlie Churn Prediction ●



Upstream dbt Health



Charlie Churn Prediction: Charlie - Scoring V1: succeeded Aug 27, 6:05 AM ET



1 - Daily: succeeded Aug 27, 5:17 AM ET



1 - Daily - Pendo Usage: succeeded Aug 27, 5:48 AM ET

Volume: 535 | Score: 0.3694



Eve Events Forecasting ●



Upstream dbt Health



Eve Events Forecasting: Eve - Model Forecasts: succeeded Aug 27, 7:00 AM ET



Eve - Future App Training: succeeded Aug 27, 4:21 AM ET



Eve - Other Level Models: succeeded Aug 27, 4:56 AM ET



1 - Daily: succeeded Aug 27, 6:26 AM ET

Volume: 0 | Score: 0.0554



Nightingale Customer Health ●



Upstream dbt Health



Nightingale Customer Health: Nightingale - Scoring V3: succeeded Aug 27, 6:20 AM ET



1 - Daily: succeeded Aug 27, 4:27 AM ET



1 - Daily - Pendo Usage: succeeded Aug 27, 6:15 AM ET



Datasync: succeeded Aug 27, 4:59 AM ET



Ward Segment B

Today ▾



Upstream dbt Health



Ward Segment B: Predict Base: succeeded Aug 27, 9:00 AM ET



1 - Daily: succeeded Aug 27, 7:20 AM ET



1 - Daily - Pendo Usage: succeeded Aug 27, 9:07 AM ET

Accuracy



Precision: 0.4715 (threshold: 0.33)



Recall: 0.5571 (threshold: 0.5)



Volume: 195 accounts scored



Ward Segment A



Upstream dbt Health



Ward Segment A: Predict Base: succeeded Aug 27, 9:00 AM ET



1 - Daily: succeeded Aug 27, 6:01 AM ET



1 - Daily - Pendo Usage: succeeded Aug 27, 7:13 AM ET

Accuracy



Precision: 0.4686 (threshold: 0.51)



Recall: 0.4442 (threshold: 0.6)



Volume: 664 accounts scored



Zeke Lead Scoring (eZQL)



Upstream dbt Health



Zeke Lead Scoring (eZQL): Zeke - Scoring V1: succeeded Aug 27, 7:55 AM ET



1 - Daily: succeeded Aug 27, 6:20 AM ET



1 - Daily - Pendo Usage: succeeded Aug 27, 5:08 AM ET



Datasync: succeeded Aug 27, 6:53 AM ET



Lindy Bustabad Today at 9:51 AM

@DS ModelGuard what enclosures need checks today?

5 replies



DS ModelGuard APP Today at 9:52 AM

Enclosure check complete — action required 🔴

- Ward Segment A 🔴: Precision 0.47 (threshold 0.51), Recall 0.44 (threshold 0.60) — both breached. Features `product_engagement_30d` (PSI 0.32) and `activity_events_30d` (PSI 0.30) both red.
- Charlie Churn Prediction 🟡: Volume down 19% vs 7-day avg (535 vs 663). Score jumped 0.25→0.37 day-over-day. Features stable.

Action: Ward Segment A needs immediate investigation — performance and feature drift are both degraded. Monitor Charlie for a second day before escalating.

Remaining models 🟢.



Lindy Bustabad Today at 9:52 AM

Tell me more about Charlie... do I need to be worried?



DS ModelGuard APP Today at 9:53 AM

Probably not. This pattern has Charlie's fingerprints all over it.

Both the Aug 3 and Jul 9 incidents looked identical — score spike, volume dip, dbt clean — and both self-resolved within 24 hours. The root cause each time was a one-day cohort of accounts with unusual renewal timing skewing the scoring population, not a real model degradation.

Today's numbers fit that same mold: volume is down 19% vs the 7-day avg, score jumped from 0.25 to 0.37, and all three upstream dbt jobs ran clean this morning. No feature drift is measured for Charlie yet (Phase 3 monitoring not built), so we can't rule it out entirely — but the upstream health being green makes a data pipeline issue unlikely.

My read: **watch it tomorrow morning**. If the score normalizes back toward 0.25 and volume recovers, it's noise again. If it doesn't, then escalate.



Lindy Bustabad Today at 9:56 AM

Any pointers on where to start triage for Ward Seg A?



DS ModelGuard APP Today at 9:57 AM

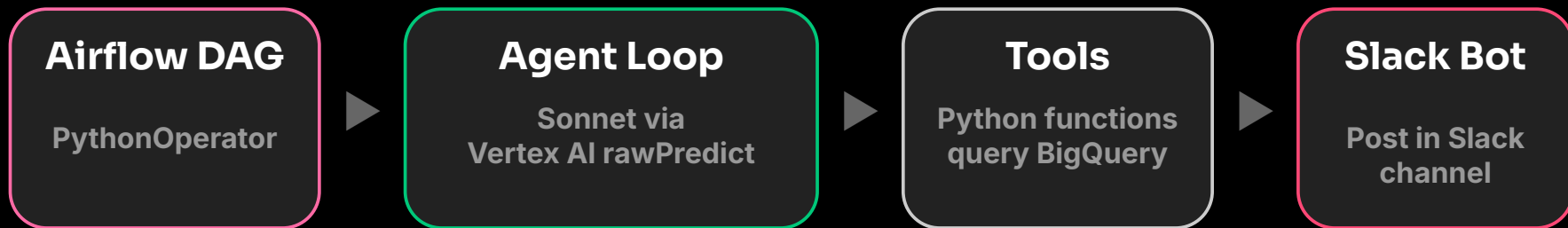
Strong pattern match here — this looks nearly identical to the **Aug 12 incident**. Same two features (`product_engagement_30d` and `activity_events_30d`) drifted, same red severity. That time the root cause was a column rename in `stg_pendo__usage_metrics` that wasn't reflected in the dbt model. Precision dropped to 0.44 then (vs. 0.47 today), and it was fixed same-day with a staging SQL patch + pipeline re-run.

Start here:

- [Daily Pipeline Job \(dbt Cloud\)](#) — check whether any upstream schema changes landed recently, especially in `stg_pendo__usage_metrics`.
- [ModelGuard Feature Drift — Ward Segment A](#) — PSI trend history for both drifting features; see if the spike is a one-day blip or building.

If you spot a schema change, the Aug 12 fix (update staging SQL, re-run the job) is the playbook.

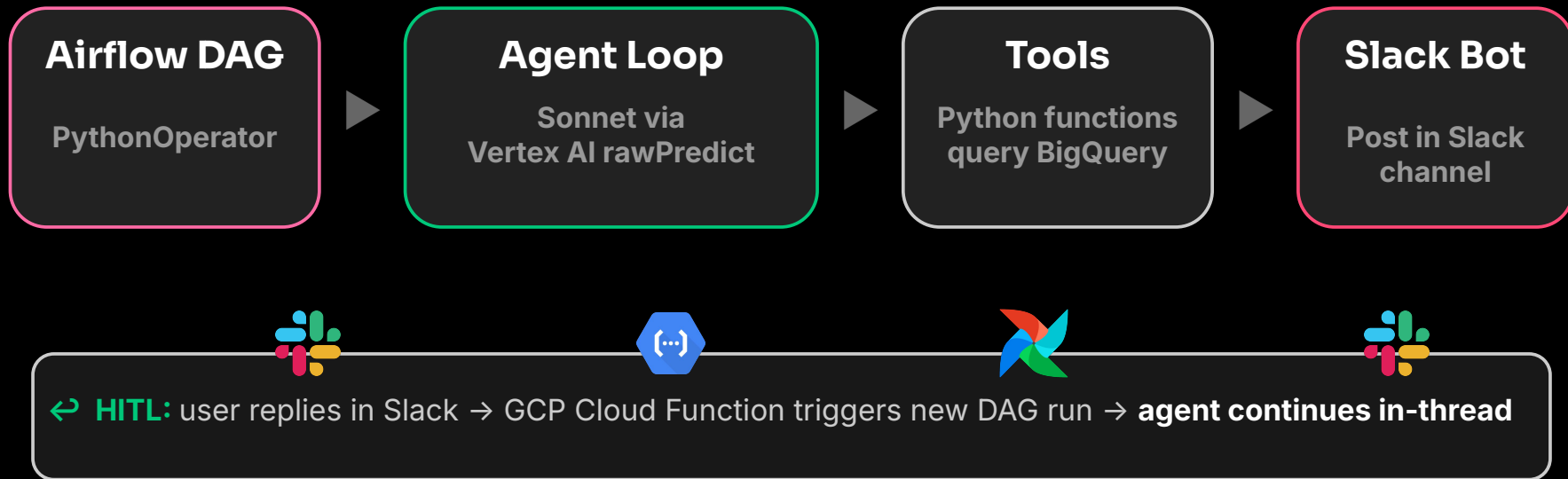
Hand-Rolled Agent on Airflow 2.x



```
{local_task_job_runner.py:123} ▶ Pre task execution logs
{modelguard_agent.py:261} INFO - LLM: I'll start by pulling the summary across all 7 models, then drill into any yellow/red ones.
{modelguard_agent.py:268} INFO - Round 1 → LLM selected 1 tool(s): get_all_models_summary
{modelguard_agent.py:281} INFO - → get_all_models_summary()
{modelguard_agent.py:261} INFO - LLM: Two models need drilling: Ward Segment A (red) and Charlie Churn Prediction (yellow). I'll pull all their signals
{modelguard_agent.py:268} INFO - Round 2 → LLM selected 4 tool(s): get_predict_model_health, get_homegrown_model_signal, get_feature_drift_status, get_fea
{modelguard_agent.py:281} INFO - → get_predict_model_health({"model_key":"ward_seg_a"})
{modelguard_agent.py:281} INFO - → get_homegrown_model_signal({"model_key":"charlie"})
{modelguard_agent.py:281} INFO - → get_feature_drift_status({"model_key":"ward_seg_a"})
{modelguard_agent.py:281} INFO - → get_feature_drift_status({"model_key":"charlie"})
{modelguard_agent.py:261} INFO - LLM: *Enclosure check complete - 1 red, 1 yellow* 🔴 • Ward Segment A 🔴: Precision 0.47 (threshold 0.51), Recall 0.51
{modelguard_agent.py:264} INFO - Round 3 → stop_reason=end_turn, no more tool calls
{python.py:240} INFO - Done. Returned value was: [{'name': 'get_all_models_summary', 'args': {}}, {'name': 'get_predict_model_health', 'args': {'model_key
{taskinstance.py:341} ▶ Post task execution logs
```

ARCHITECTURE

Hand-Rolled Agent on Airflow 2.x



The Resolution Journal

```
model: ward_seg_a
date: 2026-08-12
severity: red
root_cause: staging_schema_change
features_involved:
  - product_engagement_30d
  - activity_events_30d
summary: >
  Precision dropped to 0.44 after col rename
  in stg_pendo__usage_metrics. PSI 0.31, 0.28.
resolution: >
  Fixed staging model, re-ran pipeline.
  Recovered next morning.
resolved_in: same day
action_taken: dbt model fix + pipeline re-run
```

```
model: charlie
date: 2026-08-03
severity: yellow
root_cause: score_spike_noise
features_involved:
  - churn_risk_score
  - renewal_probability
summary: >
  Score jumped 38% DoD, volume dropped 15%.
  Features stable, dbt upstream clean.
resolution: >
  Self-resolved in 24h. Cohort of unusual
  renewal timing entering scoring population.
resolved_in: 1 day
action_taken: none - noise confirmed
```

WHAT'S NEXT

Airflow 3 and @task.agent

NOW (Airflow 2.x)

```
for round_num in range(MAX_ROUNDS + 1):
    resp = requests.post(URL, json=body)
    tool_blocks = [b for b in resp
                    if b["type"] == "tool_use"]
    if not tool_blocks:
        break
    for block in tool_blocks:
        fn = TOOLS[block["name"]]
        result = fn(**block["input"])
        messages.append(...)
```

NEXT (Airflow 3)

```
@task.agent(
    model="claude-sonnet-4-6",
    tools=[get_summary, get_health,
           get_drift, get_history],
    enable_hitl_review=True,
)
def triage(context):
    return "Check all models for today."
```

@task.agent — replaces the hand-rolled loop

enable_hitl_review — native chat UI built into Airflow

Asset watchers — trigger triage when DS ModelGuard's BQ write lands

dbt refresh tool — detect stale data → trigger re-run from the same DAG

Takeaways

- 1 You can build agentic tasks on Airflow 2.x today.**
A PythonOperator + a bounded tool-calling loop. No upgrade, no new packages.
- 2 Detection and triage are different problems.**
Deterministic checks catch problems. An agentic layer routes attention to the right one first.
- 3 Institutional memory compounds.**
A resolution journal means your triage agent gets smarter with every incident.

Questions?

Lindy Bustabad

Sr. Analytics Engineer, Pendo



Let's connect
on LinkedIn!



Taming the MLOps Zoo: Orchestrating and Monitoring Models
with Airflow

Please share feedback!